

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Mechanical Engineering) SEMESTER - 7 Winter 2025 (Regular)

Course :Bachelor of Technology (Mechanical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 7

Subject Code & Name: BTHM702 - INDUSTRIAL ENGINEERING AND MANAGEMENT

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 Fayol's principle of Unity of Direction primarily aims to:
 - A. Ensure single boss for employees
 - B. Integrate personal and organizational goals
 - C. Coordinate activities having the same objective
 - D. Balance authority with responsibility
 - 2 The primary purpose of introducing staff authority in a line organization is to:
 - A. Replace line managers
 - B. Provide specialized advice and support to line executives
 - C. Increase centralization of power
 - D. Eliminate departmentalization
 - 3 The primary objective of *merit rating* is to:
 - A. Fix employee wages
 - B. Measure job satisfaction
 - C. Evaluate employee performance systematically
 - D. Reduce absenteeism
 - 4 Under the Apprentices Act (1961), the responsibility of providing practical training mainly lies with:
 - A. Government institutions
 - B. Trade unions
 - C. Industrial establishments
 - D. Technical universities
 - 5 Service systems differ from manufacturing systems mainly due to:
 - A. High automation
 - B. Tangible output
 - C. Simultaneous production and consumption
 - D. Standardized processes

- 6 A firm competing primarily on rapid delivery and flexibility should most appropriately adopt which manufacturing system?
 - A. Continuous production
 - B. Mass production
 - C. Job shop production
 - D. Project production
- 7 Which of the following condition most limits the effectiveness of Materials Requirement Planning (MRP)?
 - A. High product variety
 - B. Inaccurate master production schedule
 - C. Low inventory turnover
 - D. Long setup times
- 8 Process-focused layouts are most suitable when production involves:
 - A. High volume and low variety
 - B. Low volume and high variety
 - C. Continuous flow
 - D. Automated assembly lines
- 9 In assembly line balancing, the theoretical minimum number of workstations is determined by:
 - A. Total task time $\div$ cycle time
 - B. Maximum elemental task time
 - C. Production volume
 - D. Plant floor area
- 10 Six Sigma differs from Total Quality Control mainly because Six Sigma emphasizes:
 - A. Employee involvement
 - B. Zero inventory
 - C. Statistical rigor and DMAIC methodology
 - D. Continuous flow production
- 11 Which technique most accurately establishes standard time without direct observation of the worker?
 - A. Stopwatch time study
 - B. Work sampling
 - C. Predetermined Time Systems (PTS)
 - D. Micromotion study
- 12 Gilbreth's micromotion study primarily utilizes:
 - A. Process charts
 - B. Therbligs
 - C. Gantt charts
 - D. Control charts

Q2. Solve the following.

- A) What is Industrial Engineering? Explain the roles of an Industrial Engineer 6
- B) What is organizing? Explain departmentation, line and staff authority, and 6 decentralization.

Q3. Solve the following.

- A) Describe performance appraisal and career planning. Explain merit rating and 6 incentive schemes.
- B) Explain the staffing process. Discuss recruitment, selection, orientation, and 6 training with reference to the Apprentice Act (1961).

- Q4. Solve Any Two of the following.
- A) Discuss different types of manufacturing systems and their characteristics. Justify their suitability for various production environments. 6
 - B) Explain in detail the Materials Requirement Planning (MRP) with the help of a neat sketch. 6
 - C) Explain various forecasting techniques used in operations planning. 6
- Q5. Solve Any Two of the following.
- A) Describe capacity planning and plant location decisions. Explain the major factors influencing these strategic choices. 6
 - B) Explain facilities layout and assembly line balancing with neat sketch. Illustrate how these improve productivity and reduce idle time. 6
 - C) Explain in brief the product and process design with the help of neat sketch. 6
- Q6. Solve Any Two of the following.
- A) Explain how JIT, Quality Circles, and Six Sigma contribute to productivity improvement under Total Quality Management? 6
 - B) What do you understand by work measurement? Explain predetermined time system (PTS). 6
 - C) Explain in detail the charting techniques and principles of motion economy with the help of neat sketch. 6

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